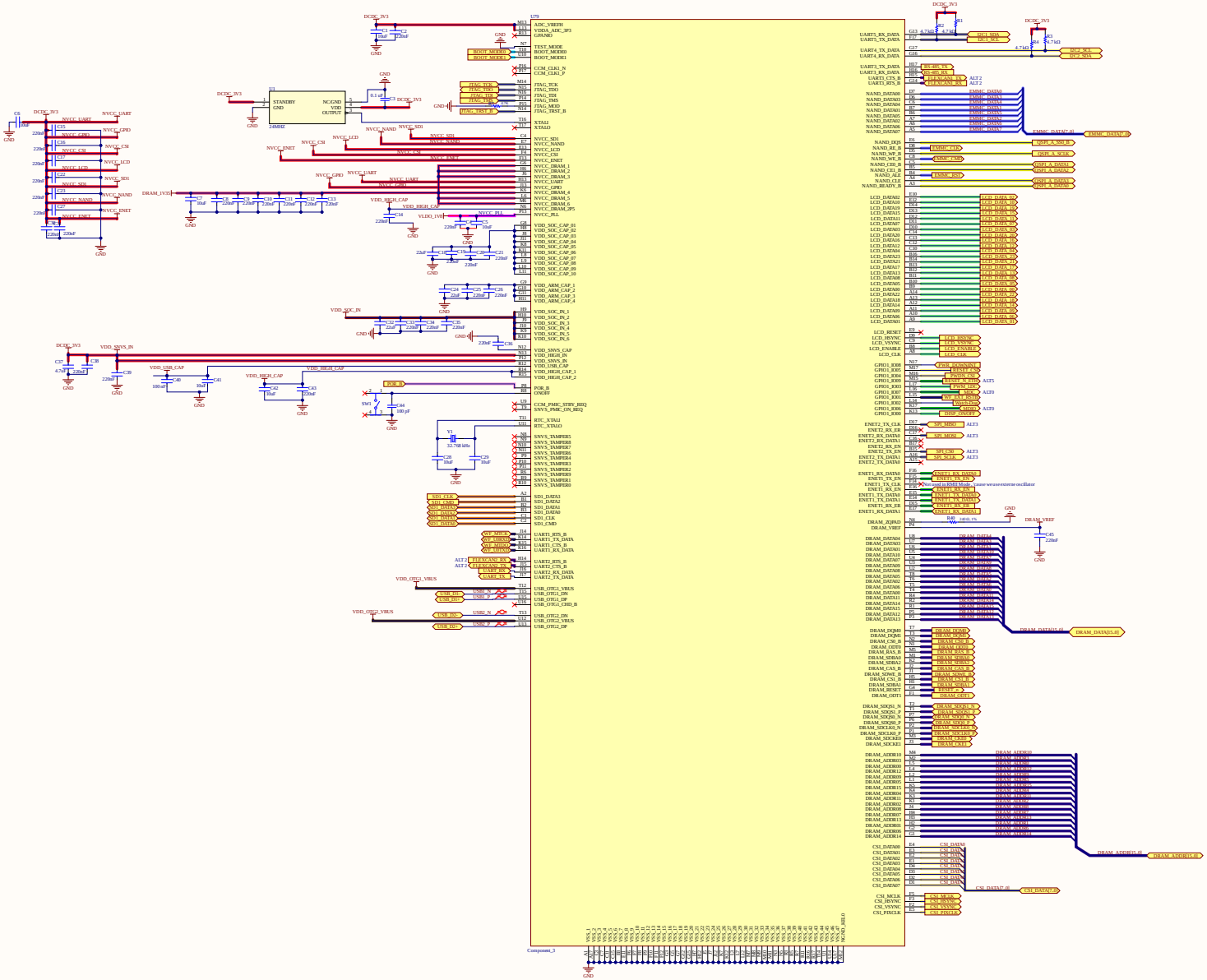
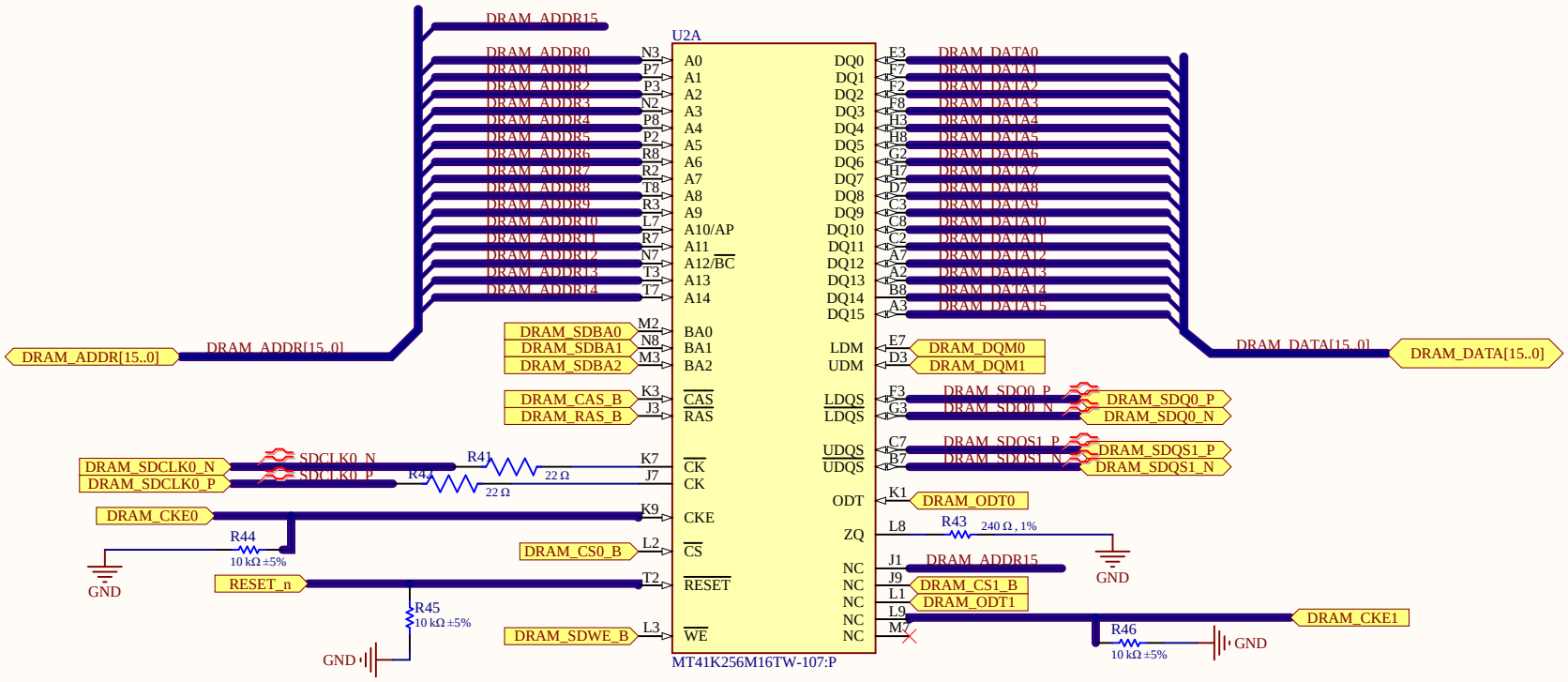
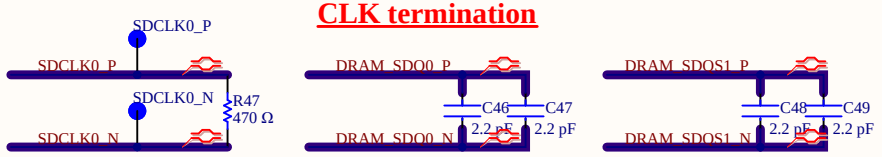
# CPU i.MX6ULL MCIMX6Y2DVM08AB



# DDR3L

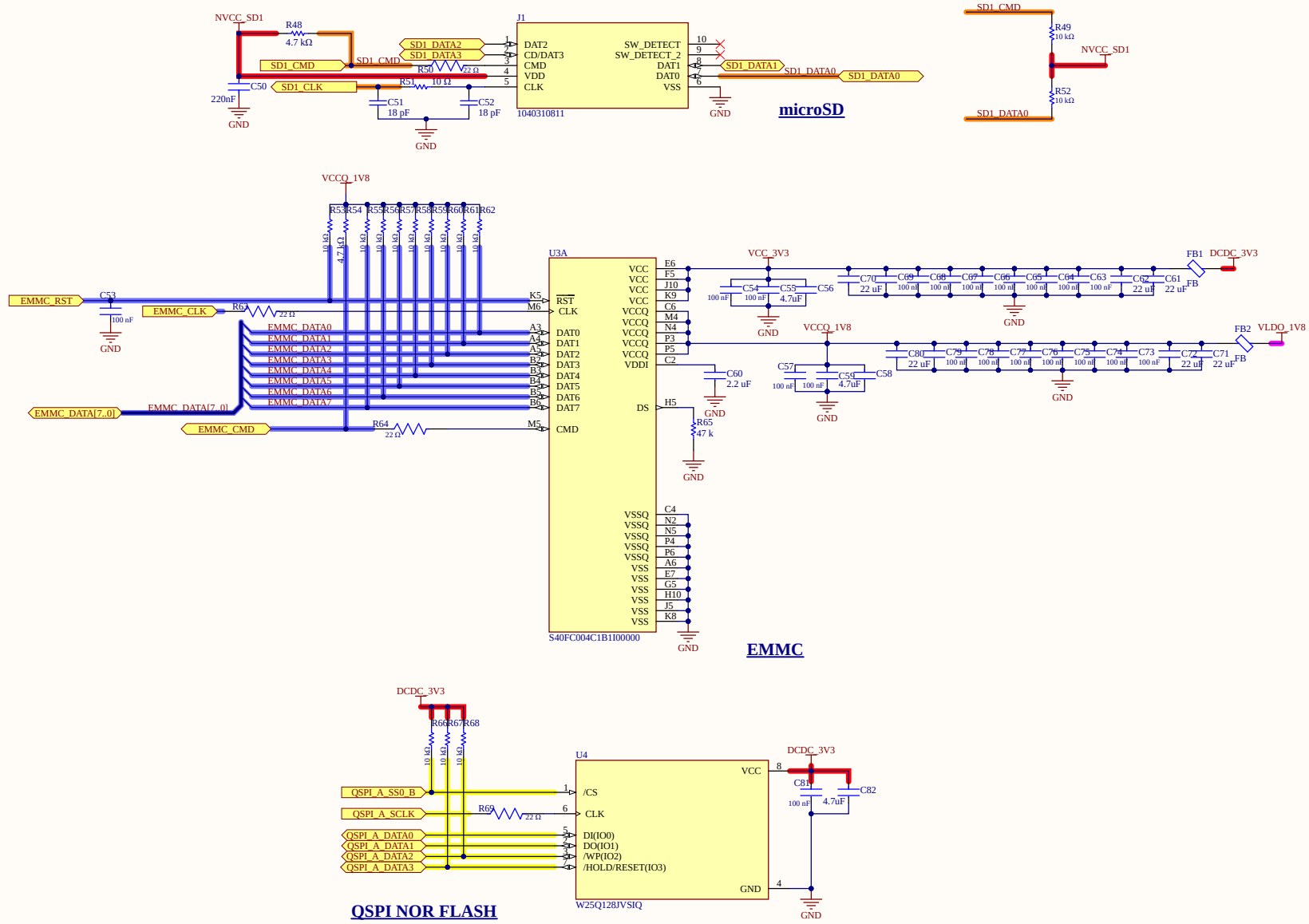



## CLK termination



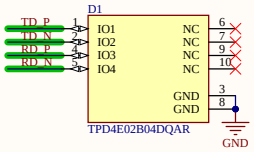

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| Title | | |
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| Date: 10/10/2026 | | Sheet of |
| File: C:\Users\..\02_DDR3L.SchDoc | | Drawn By: |

# MEMORY

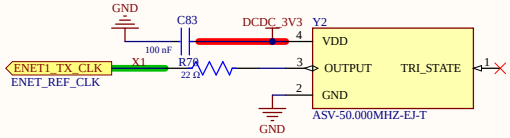



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| Date: 10/10/2026 | | Sheet of |
| File: C:\Users\..\03_Boot_Memory.SchDoc | | Drawn By: |

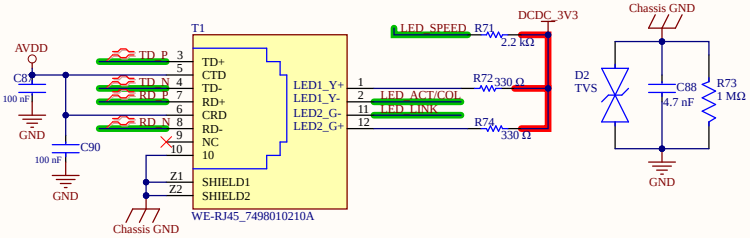
# ENET1+ RJ45



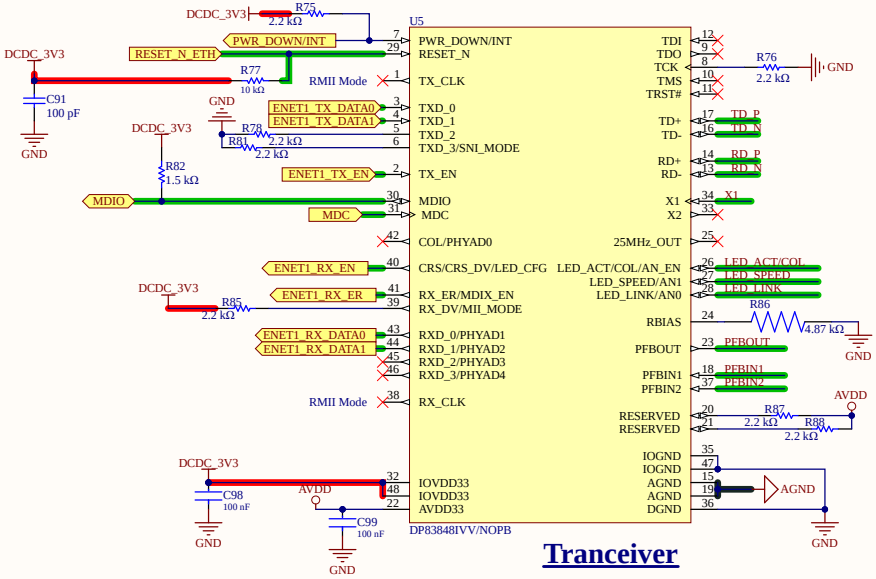

**ESD PROTECTION**



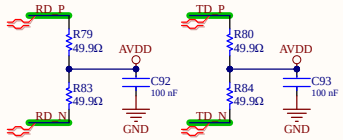

**oscillator**

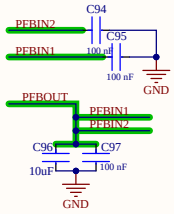



**RJ45 connector**



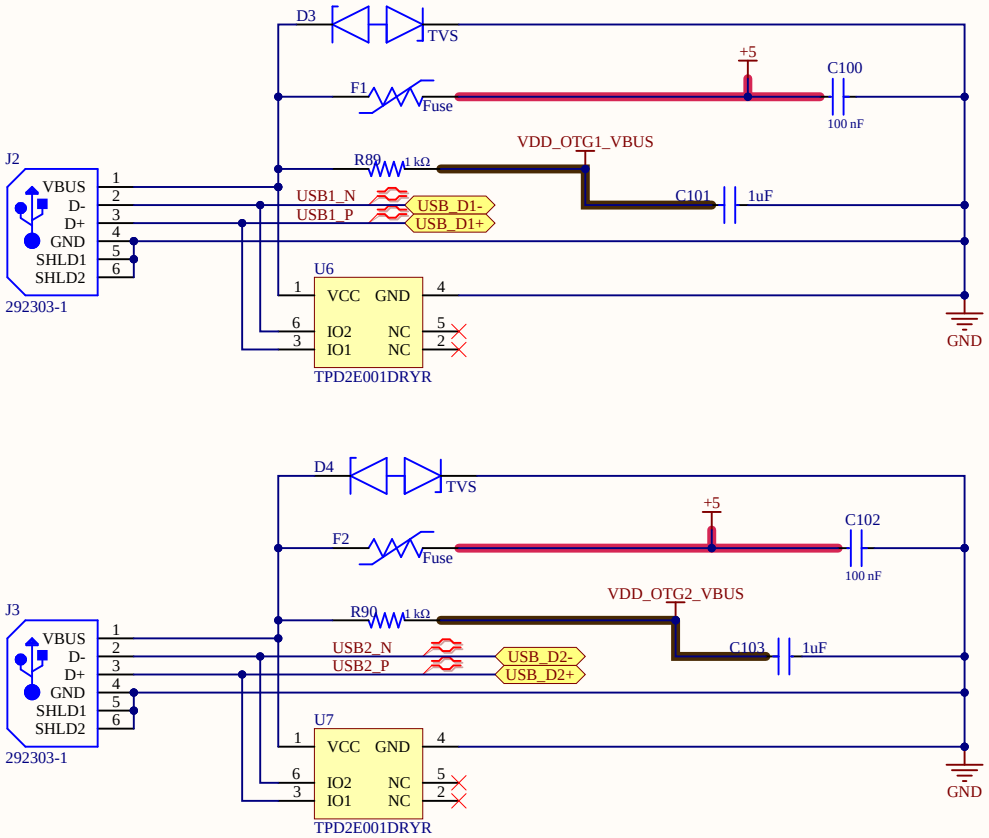

**Tranceiver**



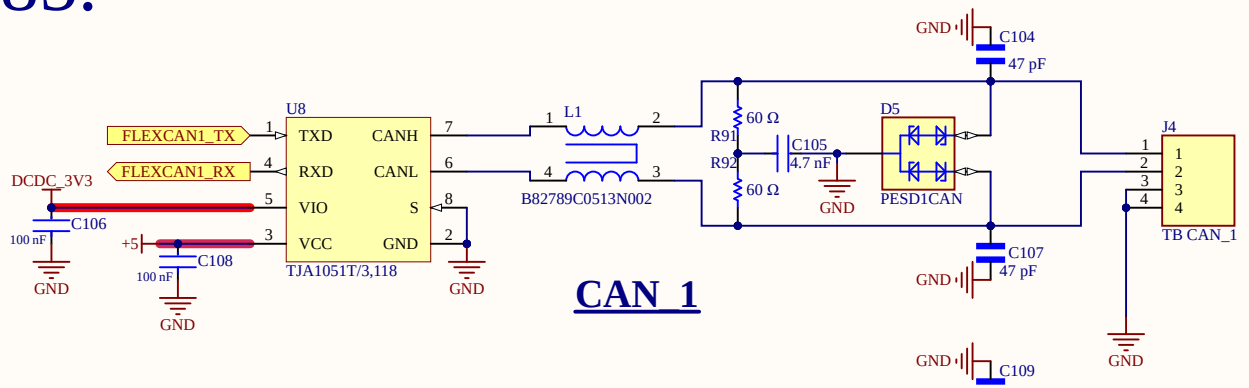

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| Title | | |
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| Date: 10/10/2026 | | Sheet of |
| File: C:\Users\..\04_Ethernet.SchDoc | | Drawn By: |

# USB-OTG

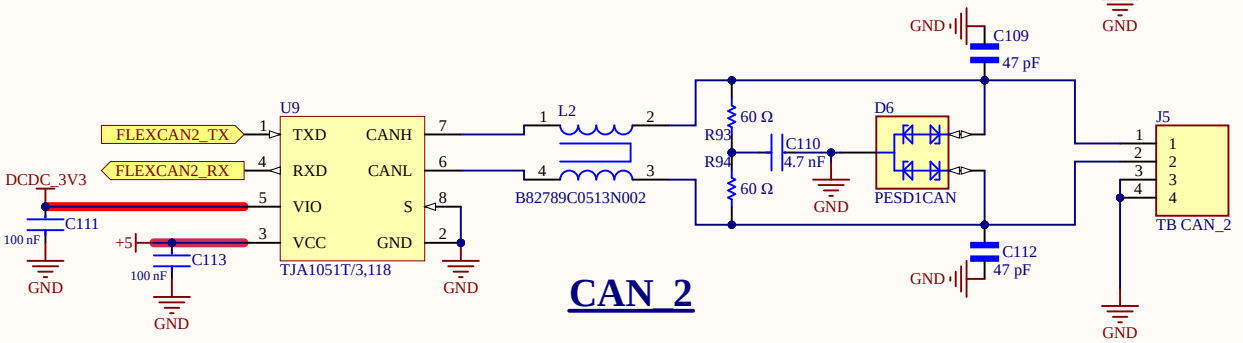



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| Title | | |
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| File: C:\Users\..\05_USB.SchDoc | | Drawn By: |

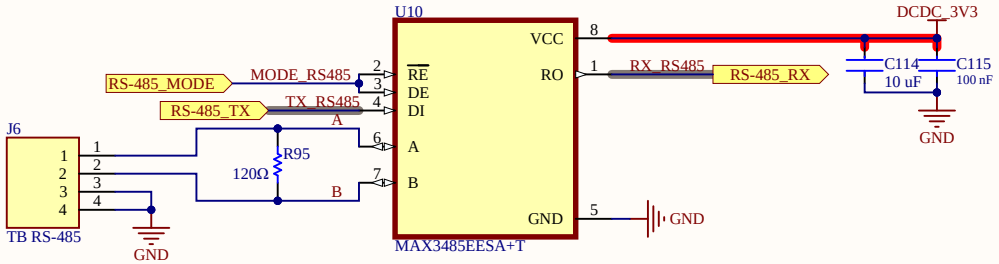
# CAN + RS-485.




## CAN_1

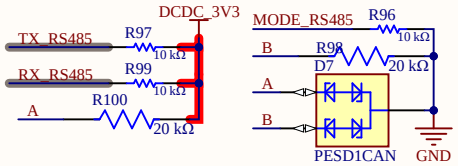



## CAN_2



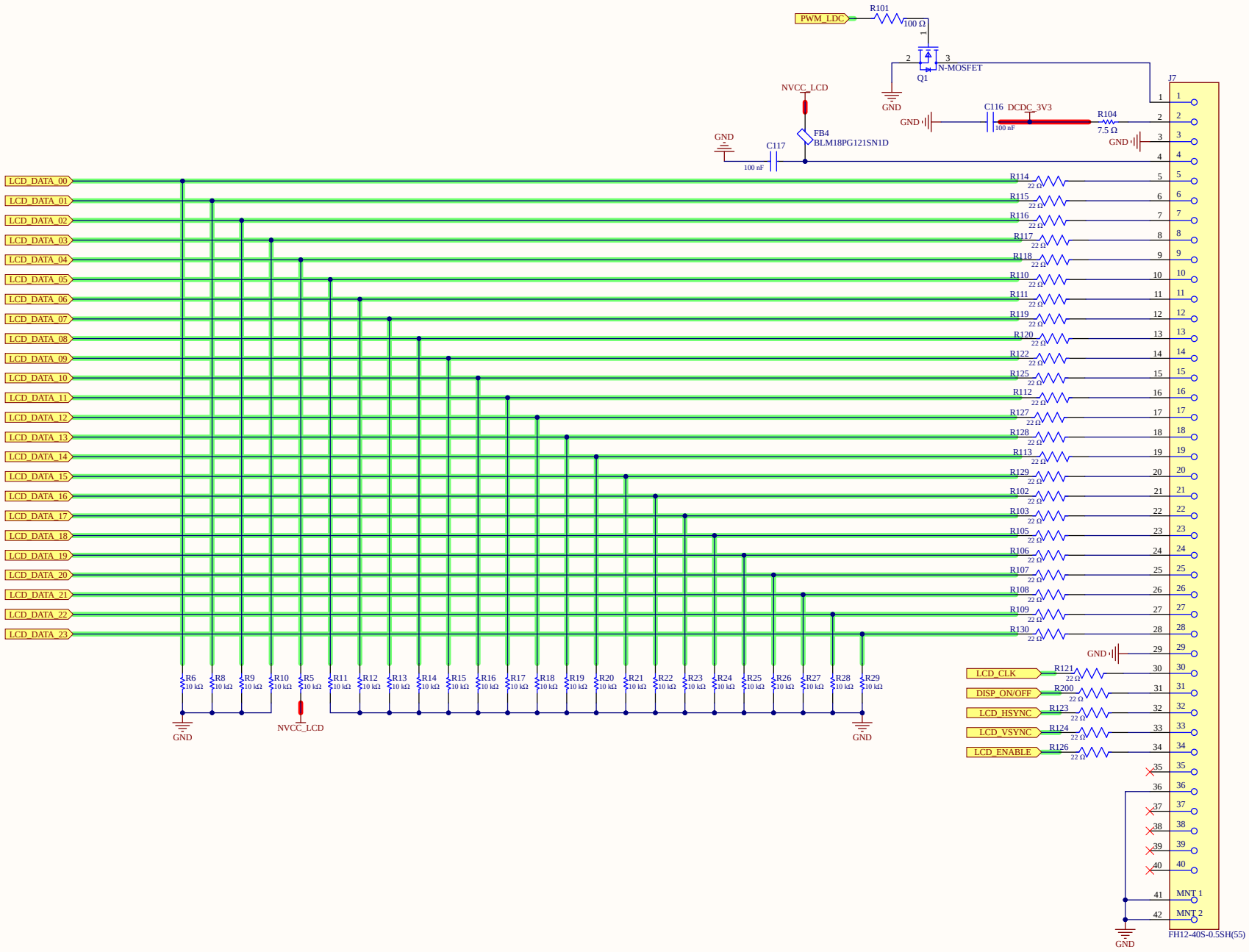

## RS-485

**MODE_RS-485 is to decide which of the transmit (DE) or the receive (RE) is ON. On reset it should be by default on Receive mode.**

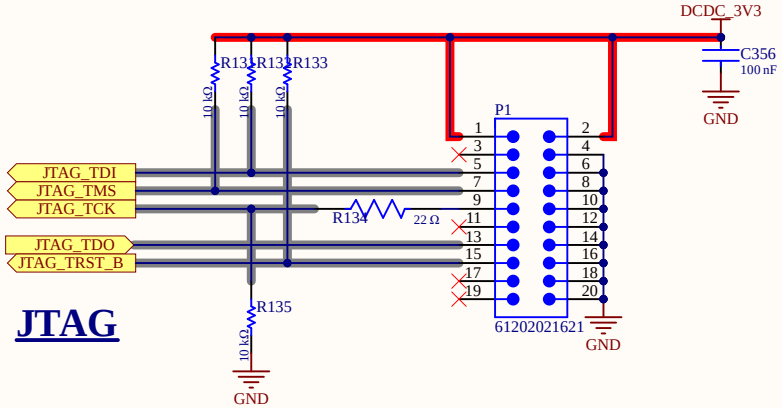


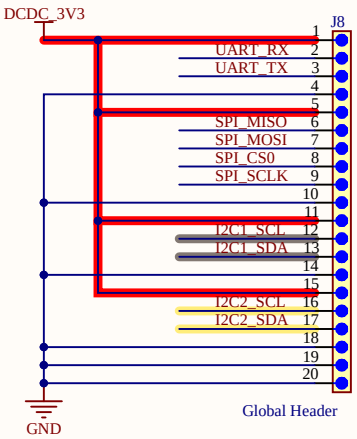

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|---|---|---|
| Title | | |
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| Date: 10/10/2026 | | Sheet of |
| File: C:\Users\..\06_CAN_RS485.SchDoc | | Drawn By: |

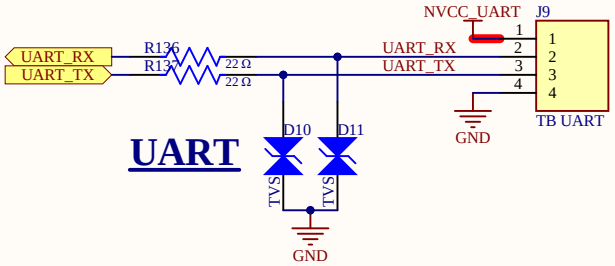
# DISPLAY + BOOT CFG

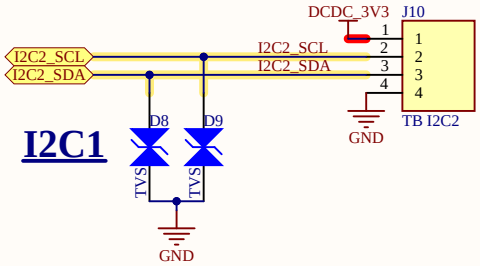


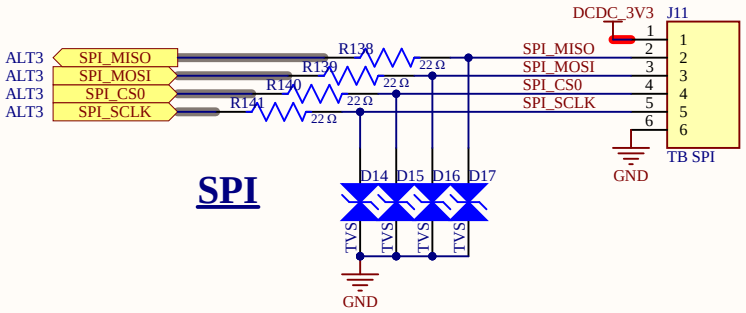
# DEBUG + Headers

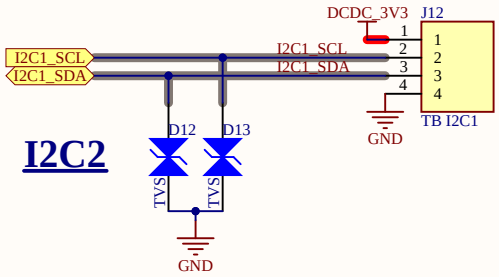


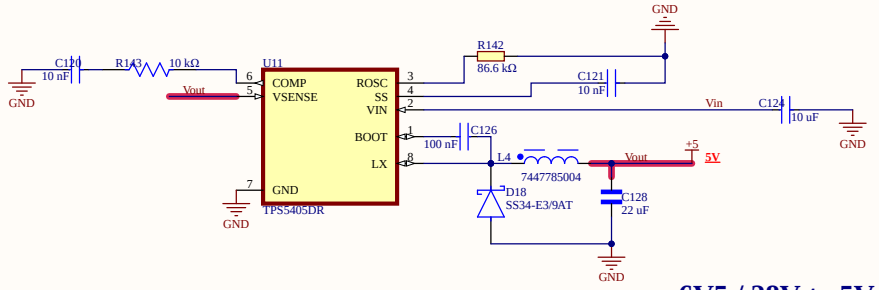



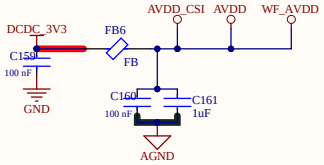




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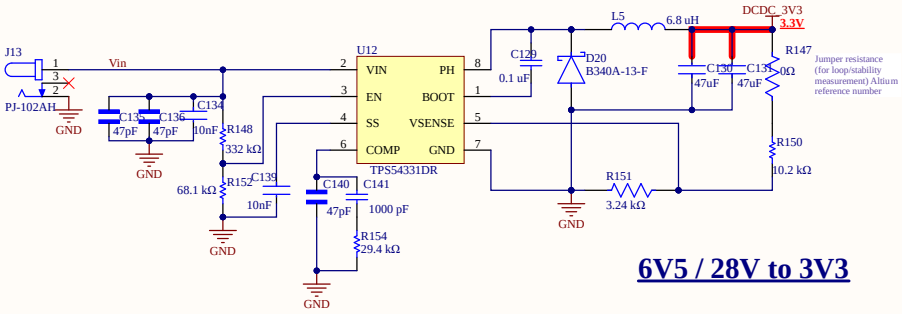
# POWER TREE



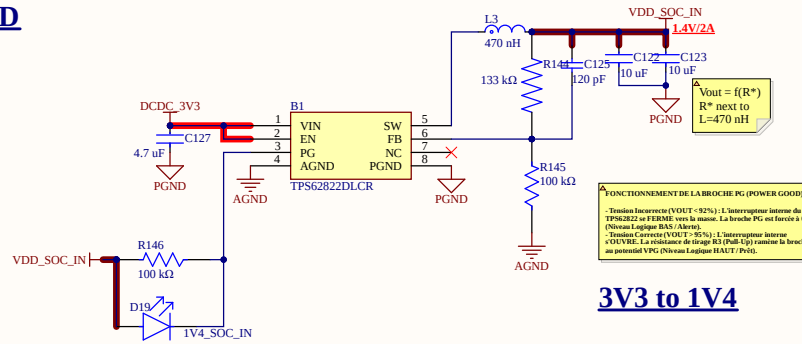

## 6V5 / 28V to 5V



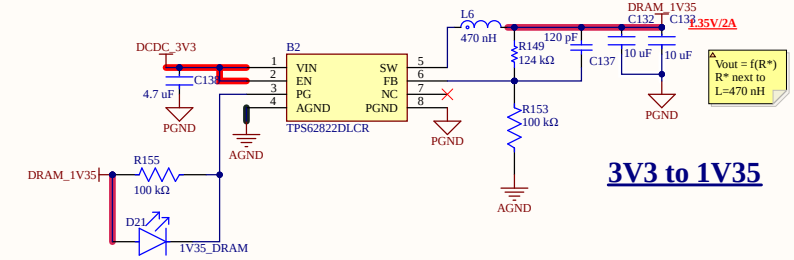

## VDD --> AVDD



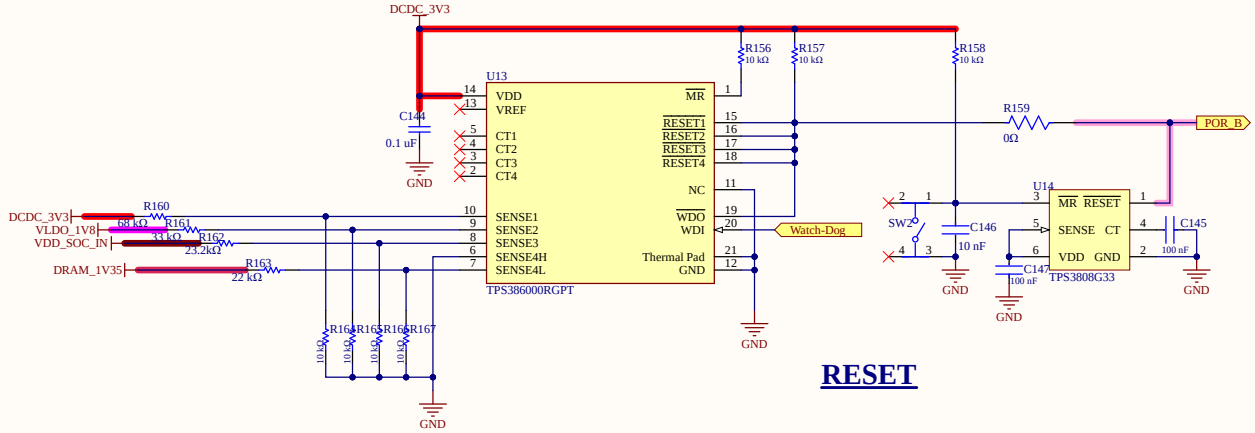

## 6V5 / 28V to 3V3



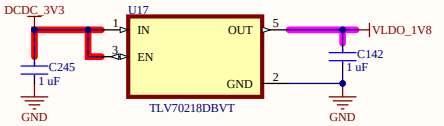

## 3V3 to 1V4



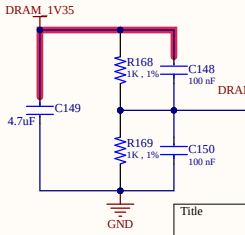

## 3V3 to 1V35




## RESET




## 3V3 to 1V8



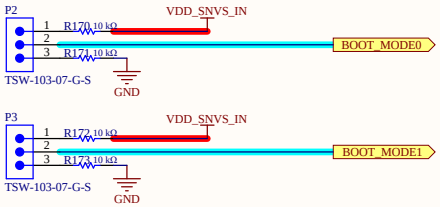

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| File: C:\Users\..\09_Power.SchDoc | | Drawn By: |

## i.MX 6ULL BOOT_MODE JUMPER CONFIGURATION

| Mode de Démarrage | Position Jumper J_BM1 | Position Jumper J_BM0 | Code Binaire | Utilisation |
|---|---|---|---|---|
| Internal Boot | Pin 1-2 (Lié à 3.3V) | Pin 2-3 (Lié à GND) | 1 0 | Mode **Standard** / Production (QSPI NOR) |
| Serial Downloader | Pin 2-3 (Lié à GND) | Pin 1-2 (Lié à 3.3V) | 0 1 | Flashing usine via USB OTG (uuu tool) |
| Boot From Fuses | Pin 2-3 (GND) | Pin 2-3 (GND) | 0 0 | Boot selon la configuration eFuse |

## BOOT MODE

BOOT MODE SELECTION
00 — Boot From Fuses : Not used in our SBC prototype.
01 — Serial Downloader : USB recovery / flashing mode.
10 — Internal Boot : Normal boot from eMMC using the hardware boot configuration.
11 — Not used.




## FUSE MAP <Default: QSPI BOOT>

| TYPE | BOOT_CFG1[7] | BOOT_CFG1[6] | BOOT_CFG1[5] | BOOT_CFG1[4] | BOOT_CFG1[3] | BOOT_CFG1[2] | BOOT_CFG1[1] | BOOT_CFG1[0] |
|---|---|---|---|---|---|---|---|---|
| | 0/1 | 0/1 | 0/1 | 1 | 0 | 0 | 0 | 0 |
| QSPI | 0 | 0 | 0 | 1 | Reserved | DDRSMP: "000" - Default "001-111" | | |

**0x10**

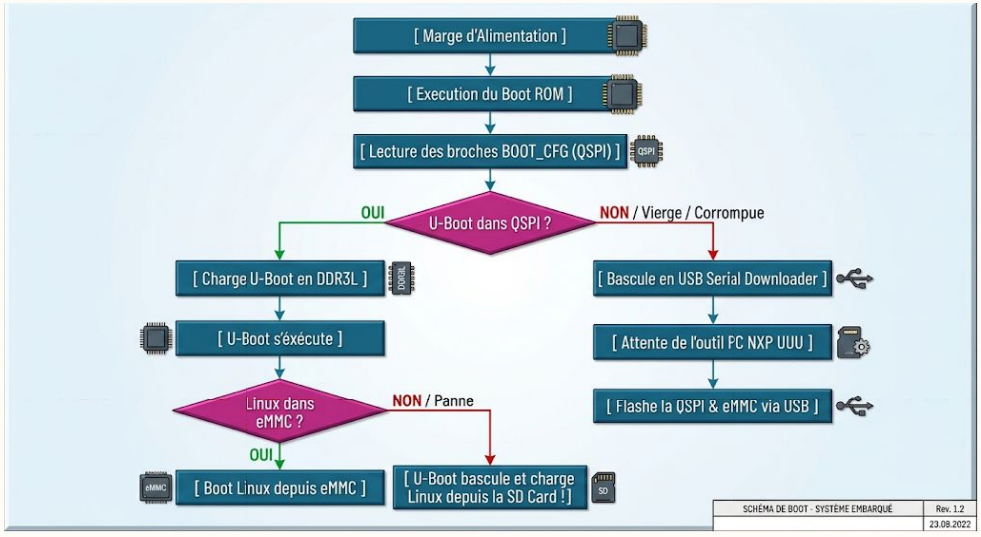
| TYPE | BOOT_CFG2[7] | BOOT_CFG2[6] | BOOT_CFG2[5] | BOOT_CFG2[4] | BOOT_CFG2[3] | BOOT_CFG2[2] | BOOT_CFG2[1] | BOOT_CFG2[0] |
|---|---|---|---|---|---|---|---|---|
| QSPI | Reserved | HSPHS: Half Speed Phase Selection 0 - select sampling at non-inverted clock 1 - select sampling at inverted clock | HSDLY: Half Speed Delay selection 0 - one clock delay 1 - two clock delay | SPHS: Full Speed Phase Selection 0 - select sampling at non-inverted clock 1 - select sampling at inverted clock | FSDLY: Full Speed Delay selection 0 - one clock delay 1 - two clock delay | Boot Frequencies 000 - 396 MHz 001 - 528 / 396 MHz 010 - 528 / 450 MHz 011 - 200 / 100 MHz | Reserved | Reserved |

**0x00**

| TYPE | BOOT_CFG4[7] | BOOT_CFG4[6] | BOOT_CFG4[5] | BOOT_CFG4[4] | BOOT_CFG4[3] | BOOT_CFG4[2] | BOOT_CFG4[1] | BOOT_CFG4[0] |
|---|---|---|---|---|---|---|---|---|
| 0x450 | Infinit-Loop (Debug USE only) 0 - Disable 1- Enable | EEPROM Recovery Enable '0' - Disabled '1' - Enabled | CS select (SPI only): 00 - CS#0 (default) 01 - CS#1 10 - CS#2 11 - CS#3 | | SPI Addressing: 0 - 2-bytes (16-bit) 1 - 3-bytes (24-bit) | | Port Select: 000 - eCSPI1 001 - eCSPI2 010 - eCSPI3 011 - eCSPI4 100 - Reserved 101 - Reserved 110 - Reserved 111 - Reserved | |
| 0x460 | L2_HW_INVALIDATE _DISABLE | Reserved | FORCE_COLD_BOOT (Reflected in SBMR2) | BT_FUSE_SEL | DIR_BT_DIS | Reserved | SEC_CONFIG[1] | Reserved |
| 0x460 | Reserved (DDR3 config options) | | | | | | | |
| 0x460 | JTAG_SMODE[1:0] | WDOG_ENABLE '0' - Disabled '1' - Enabled | SJC_DISABLE | Reserved | Reserved | Reserved | Reserved | Reserved |
| 0x460 | Reserved | Reserved | Reserved | TZASC_ENABLE | JTAG_HEO | KTE | Reserved | DLL_ENABLE 0 - Disable DLL for SD/eMMC 1 - Enable DLL for SD/Emmc |
| 0x470 | DLL Override: 0 - DLL Slave Mode for SD/eMMC 1 - DLL Override Mode for SD/eMMC | Reserved | SD2 VOLTAGE SELECTION 0 - 3.3V 1 - 1.8V | Reserved | Disable SDMMC Manufacture mode 0 - Enable 1 - Disable | L1 I-Cache DISABLE | BT_MMU _DISABLE | Override Pad Settings (using PAD_SETTINGS value) |
| 0x470 | Reserved for unexpected requirements | eMMC 4.4 - RESET TO PRE-IDLE STATE | Override HYS bit for SD/MMC pads | USDHC_PAD_PULL_DOWN 0 - no action 1 - pull down | ENABLE_EMMC_22K_PULLUP 0 - 47K pullup 1 - 22K pullup | ADD_DS_SET_GPR1_16 0 - Set 1 - Don't set | USDHC_IOMUX_SION_BIT_ENABLE 0 - Disable 1 - Enable | USDHC_IOMUX SRE Enable 0 - Disable 1 - Enable |
| 0x470 | USDHC_CMD_OE_PRE_EN (SD/MMC debug) | LPB_BOOT [Core / DDR- Bus] '00' - LPB Disable '01' - 1 GPIO Ref Freq '10' - Div by2 '11' - Div by 4 | | BT_LPB_POLARITY (GPIO polarity) | POWER_MNG_CFG (LDO's DCDC's) (Reserved - NOT USED) | | | |
| 0x470 | Override NAND Pad Settings (using PAD_SETTINGS value) | MMC_DLL_DLY[6:0] Delay target for SD/eMMC DLL, it is applied to slave mode target delay or override mode target delay depends on DLL Override fuse bit value. | | | | | | |

**0x00**

## LCD BOOT CONFIG TABLE

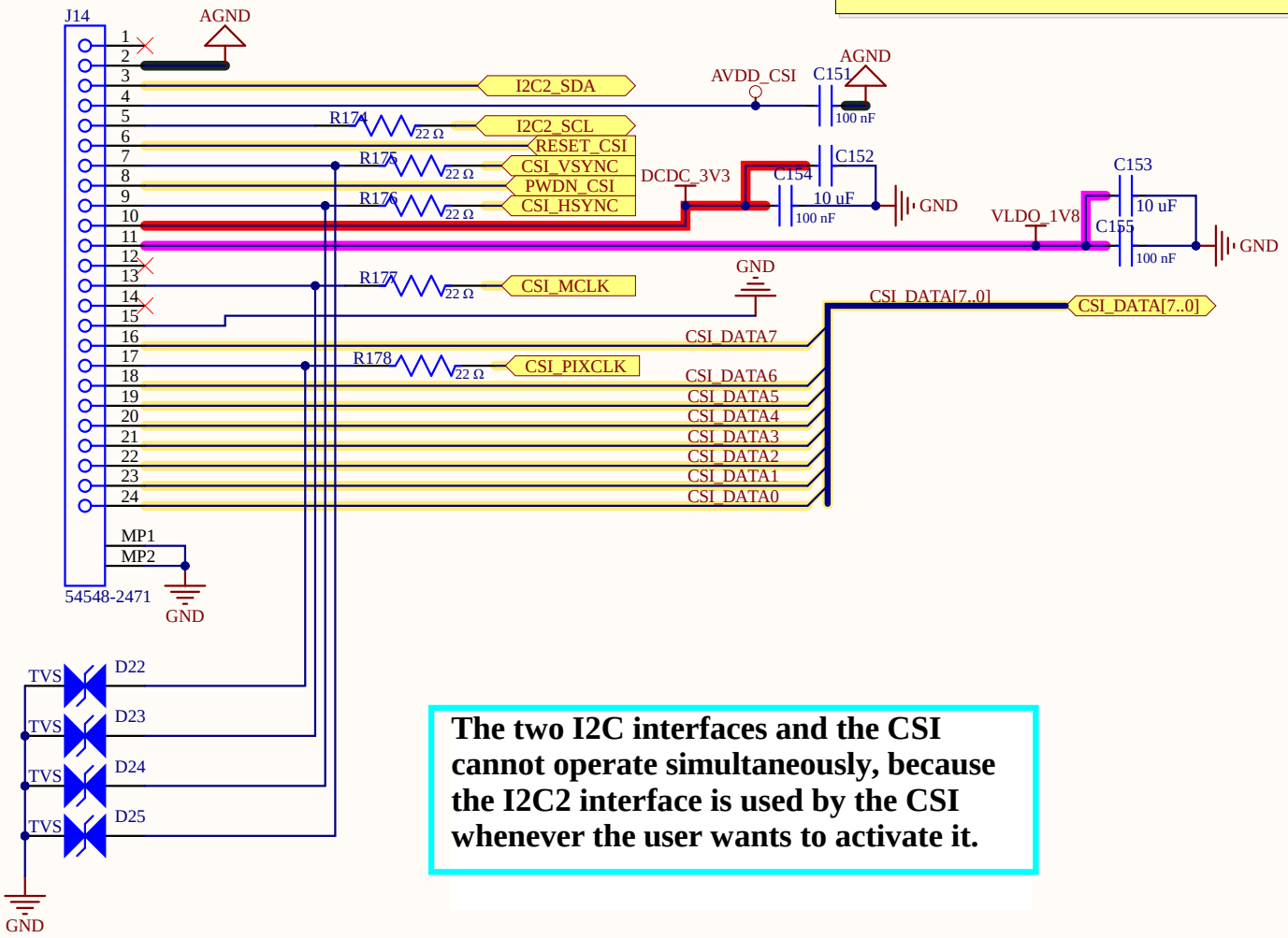



## BOOT ARCHITECTURE

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| Title | | |
| Size A3 | Number | Revision |
| Date: 10/10/2026 | | Sheet of |
| File: C:\Users\..\Boot_Cfg.SchDoc | | Drawn By: |

# CAMERA

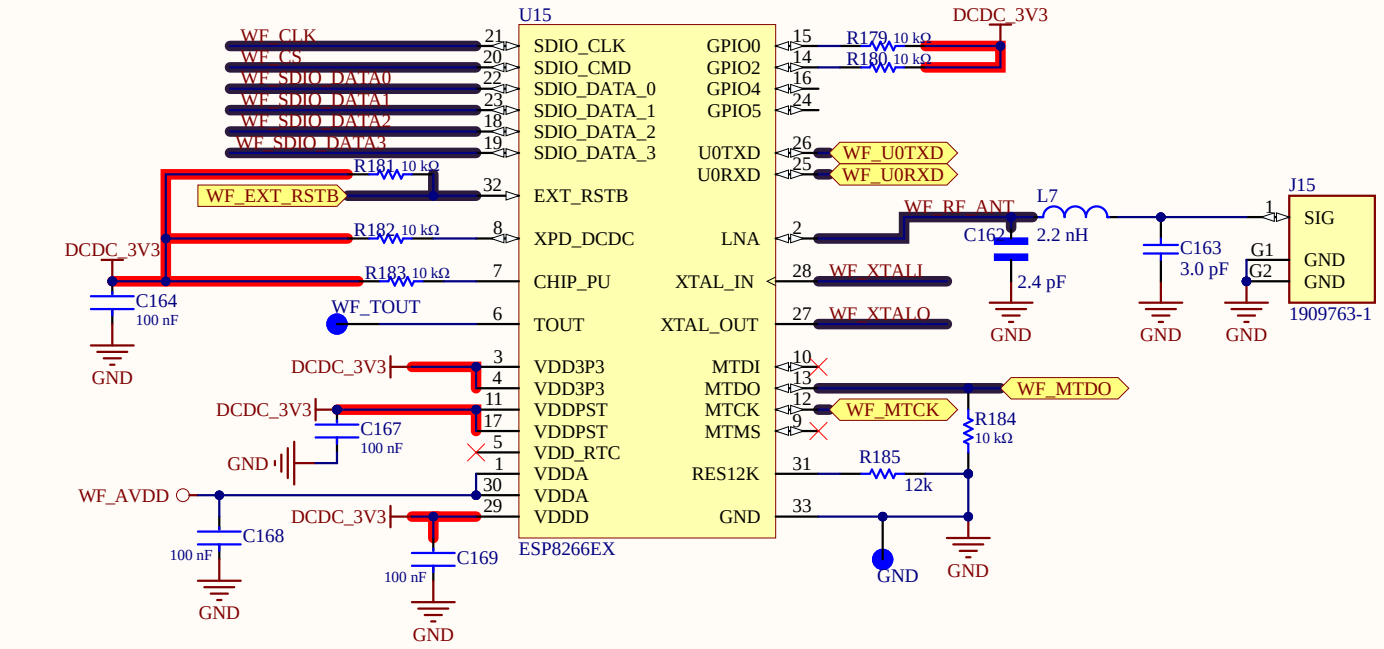
**Placer la résistance de 22 Ω sur la ligne CSI_MCLK au plus près du i.MX 6ULL. Pour les signaux CSI_PIXCLK, CSI_HSYNC et CSI_VSYNC, placer les résistances de 22 Ω au plus près de la source du signal.**

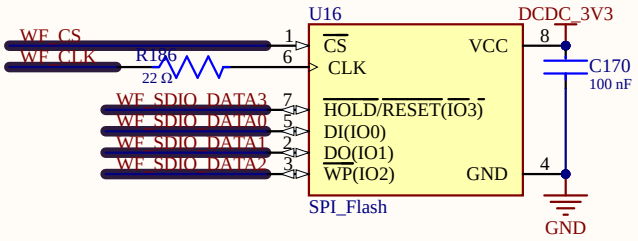


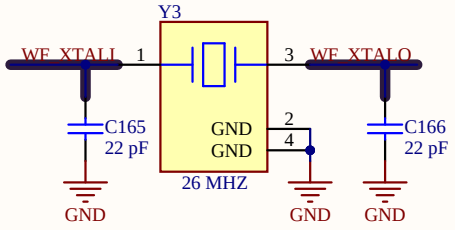

**The two I2C interfaces and the CSI cannot operate simultaneously, because the I2C2 interface is used by the CSI whenever the user wants to activate it.**

| | | |
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| Date: 10/10/2026 | | Sheet of |
| File: C:\Users\..\CSI.SchDoc | | Drawn By: |

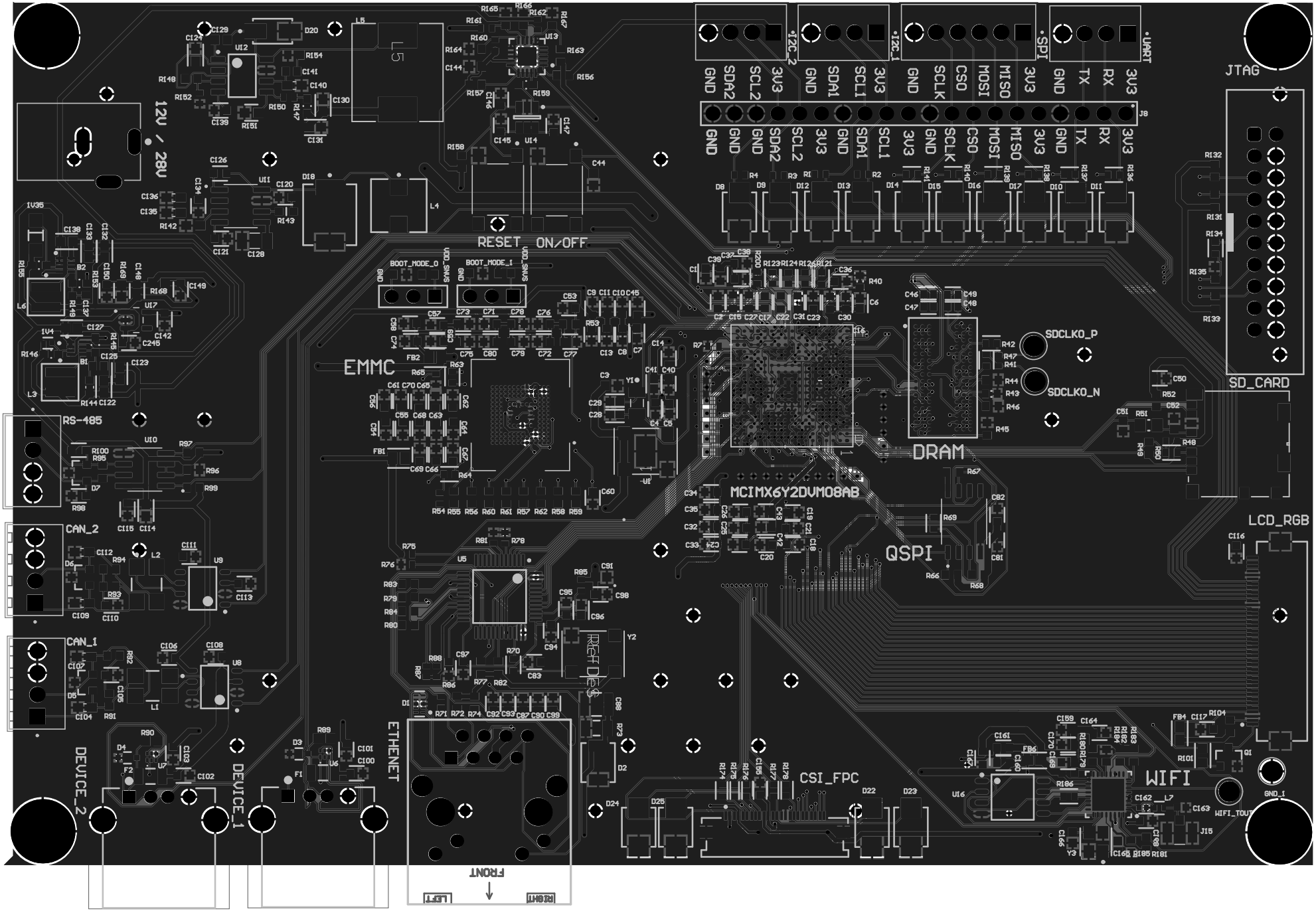
# WIFI




| | | |
|---|---|---|
| Title | | |
| Size: A | Number: | Revision: |
| Date: 10/10/2026 | | Sheet of |
| File: C:\Users\..\WIFI.SchDoc | | Drawn By: |


12V / 28V
RESET
ON/OFF
EMMC
DRAM
MCIMX6Y2DVM08AB
QSPI
SDCLKO_P
SDCLKO_N
SD_CARD
JTAG
LCD_RGB
RS-485
CAN_2
CAN_1
DEVICE_2
DEVICE_1
ETHENET
CSI_FPC
WIFI
WIFI_TOUT
GND_1
BOOT_MODE_0
BOOT_MODE_1
FRONT
LEFT
RIGHT
J8
GND
SDA2
SCL2
3V3
SDA1
SCL1
SCLK
CSO
MOSI
MISO
TX
RX
I2C_2
I2C_1
SPI
UART